

CyberShield Analytics

Cybersecurity Threat Analytics & Risk Prediction Dashboard

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Project Report

This report presents the project concept, workflow, dashboard interface design and machine-learning evaluation approach.

UI note: Screens are illustrative mockups. Values are placeholders, not actual dataset results.

1. Problem Statement

Convert raw cybersecurity records into KPIs, trends, attack patterns, risk signals and recommended actions.

2. Objectives

Clean data, identify KPIs, analyze attack categories/trends, surface risk signals, build an interactive dashboard and train a classifier when a suitable target is present.

3. Dataset

Proposed source: Kaggle Multi-Type Network Attack Detection Dataset. Verify it differs from the dataset used in the masterclass. Source: <https://www.kaggle.com/datasets/zoya77/multi-type-network-attack-detection-dataset>

4. Methodology

Raw Data → Cleaning → EDA → KPI Selection → Visualization → Risk/Opportunity Analysis → Optional ML → Actionable Insights.

5. Dashboard Features

Executive Overview: record counts and distributions. Threat Analysis: attack classes and correlations. Risk Analysis: severity/data-quality signals and recommendations. ML Prediction: Random Forest metrics and confusion matrix.

6. KPIs

Total records, attack/class distribution, severe events when available, binary attack rate where applicable, missing cells, duplicate rows and runtime model metrics.

7. Machine Learning

Preprocess numeric/categorical features, train a Random Forest model and report held-out accuracy, precision, recall, F1 and confusion matrix. Actual metrics are generated after loading the dataset.

8. Recommendations

Prioritize critical incidents, investigate frequent attack classes, monitor useful features, review false positives/negatives and periodically retrain models.

9. Limitations

Public datasets may not represent production networks. Class imbalance, leakage and outdated patterns can affect results. This prototype is decision support, not a production SIEM.

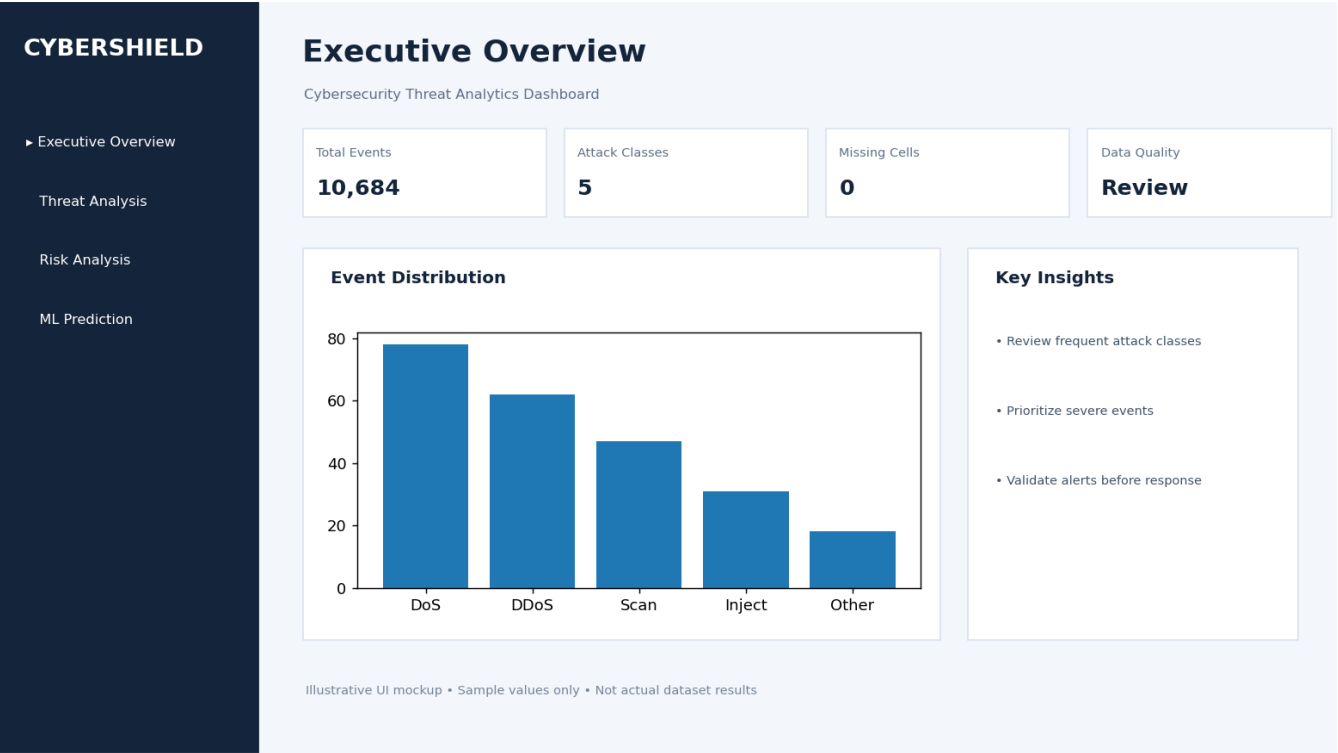
10. Conclusion

The project demonstrates a cybersecurity data-to-decision workflow focused on meaningful KPIs, trends, risk signals and actions.

11. User Interface (UI) Design

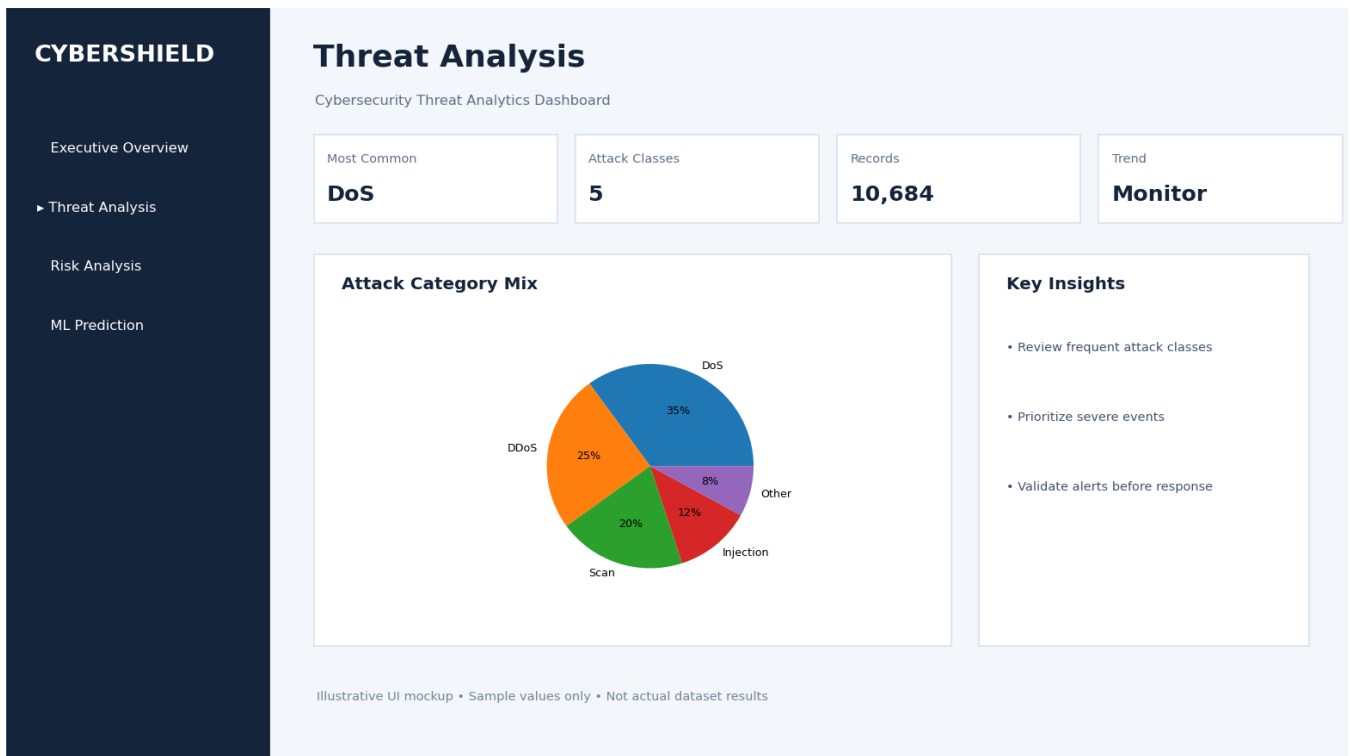
The following screens illustrate the intended dashboard. Values are placeholders for design communication only.

Executive Overview



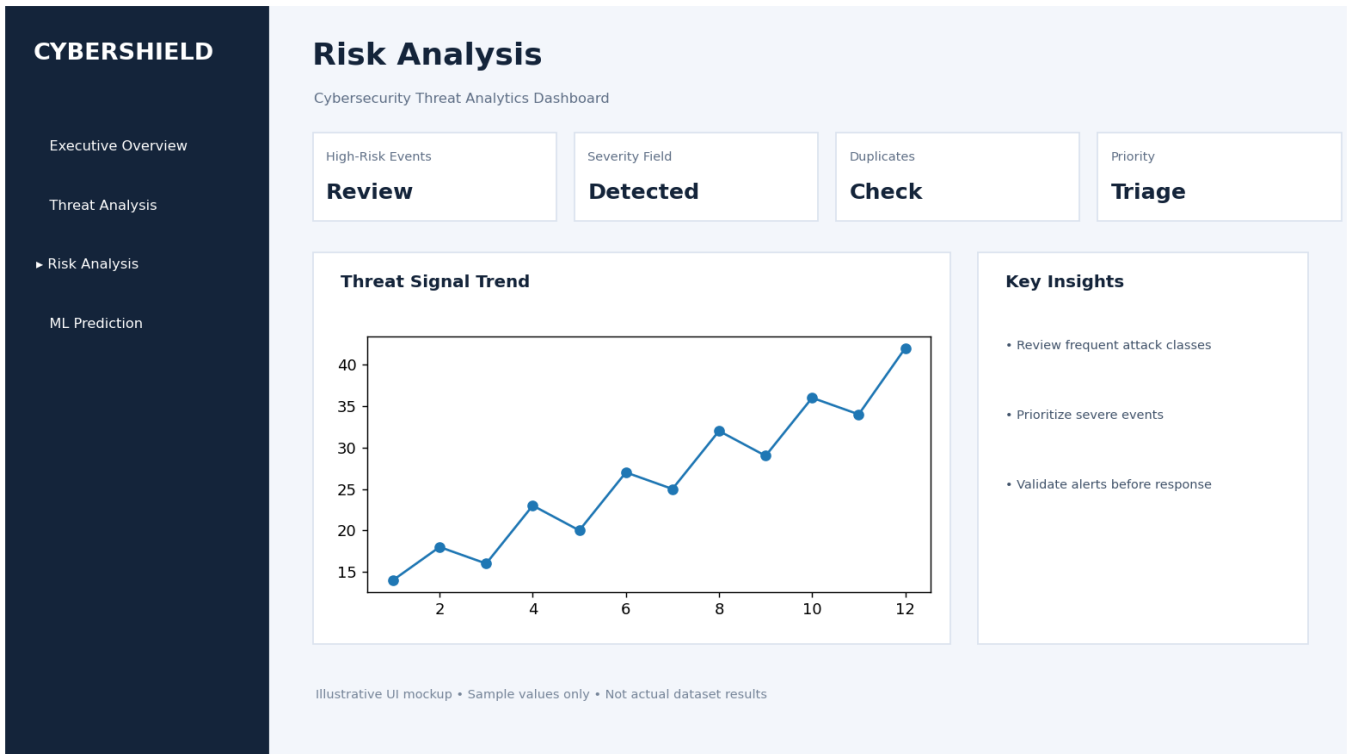
Illustrative mockup — replace with actual app screenshots after running the dataset.

Threat Analysis



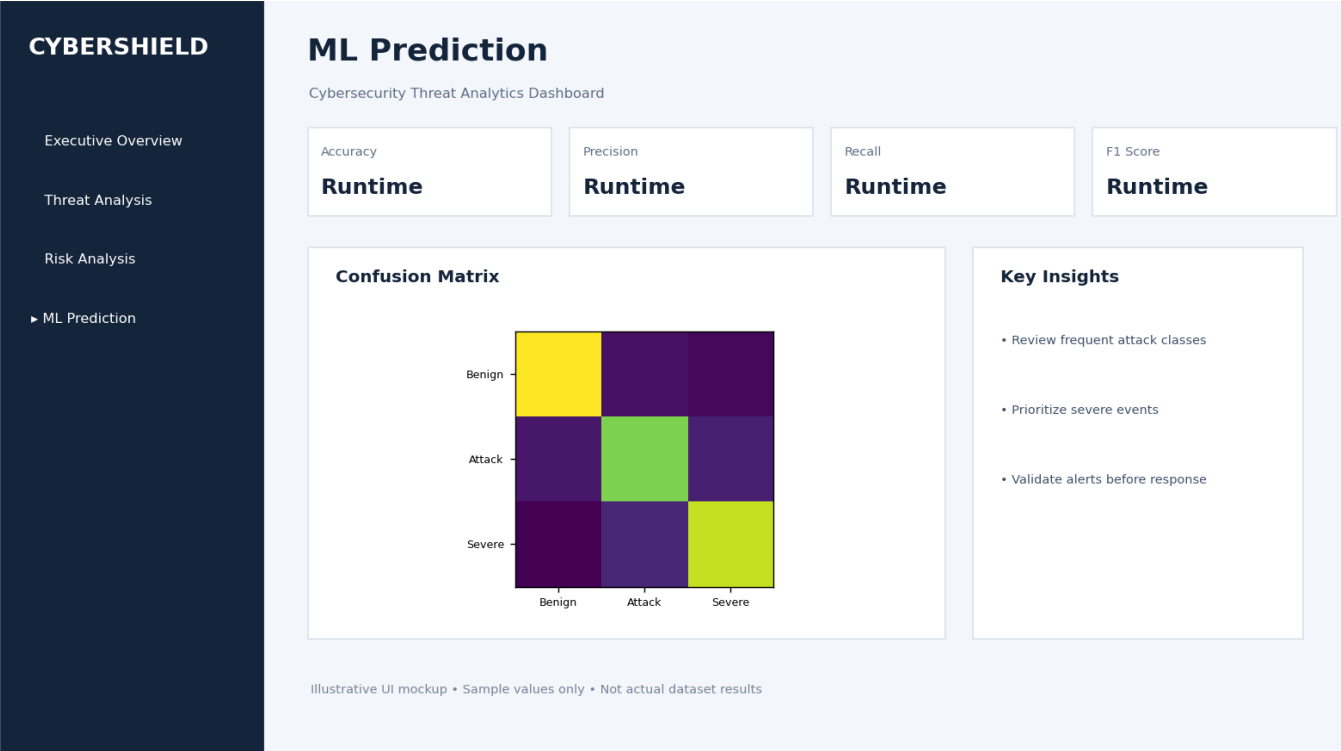
Illustrative mockup — replace with actual app screenshots after running the dataset.

Risk Analysis



Illustrative mockup — replace with actual app screenshots after running the dataset.

ML Prediction



Illustrative mockup — replace with actual app screenshots after running the dataset.

12. Deliverables & Execution

Deliverables: cybersecurity_dashboard.py, requirements.txt, README.md and this PDF report.

Run: pip install -r requirements.txt, then streamlit run cybersecurity_dashboard.py. Upload the selected CSV from the sidebar.